Exelon Generation.

10 CFR 50.73

LG-18-067
June 4, 2018

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 1
Renewed Facility Operating License No. NPF-39
NRC Docket No. 50-352

Subject: LER 2018-002-00, Primary Containment Isolation Valve Failed to Fully Close Resulting in a Condition Prohibited by TS

Enclosed is a Licensee Event Report (LER) which addresses a Primary Containment Isolation Valve (PCIV) being inoperable, resulting in a condition prohibited by Technical Specifications (TS) at Limerick Generating Station, Unit 1.

This LER is being submitted pursuant to the requirements of 10 CFR 50.73(a)(2)(i)(B), Operation or Condition Prohibited by TS.

There are no commitments contained in this letter.

If you have any questions, please contact Robert B. Dickinson at (610) 718-3400.

Respectfully,

FOR Richard W. Libra
Vice President – Limerick Generating Station
Exelon Generation Company, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, Limerick Generating Station

NRC FORM 366 (02-2018)

U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

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4. Title

Primary Containment Isolation Valve Failed to Fully Close Resulting in Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
04	03	2018	2018	- 002	- 00	06	04	2018	Facility Name	Docket Number

9. Operating Mode: 4

10. Power Level: 000

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Robert B. Dickinson, Manager – Regulatory Assurance	(610) 718-3400

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
E	BO	ISV	B580	Y					

14. Supplemental Report Expected

☐ **Yes** (If yes, complete 15. Expected Submission Date) ☒ **No**

15. Expected Submission Date	Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

During the performance of reverse flushes of the Shutdown Cooling Piping for the refuel outage, the 1A Low Pressure Coolant Injection (LPCI) Outboard Injection Primary Containment Isolation Valve (PCIV) failed to fully close. The valve was manually closed and declared inoperable. Troubleshooting identified corrosion on the torque switch prevented the electrical and mechanical switches from closing on a demand signal.

It was determined that the PCIV was inoperable after 9/8/17 when moisture from a nearby steam leak penetrated the motor operator limit switch compartment causing a ground in the motor operator. After the ground was received the valve was declared inoperable, actions were taken to clear the ground, and the valve was returned to an operable status. Though the PCIV was closed from 9/8/17 to 3/27/18, the valve was not deenergized per Technical Specification (TS) and the PCIV is required to close to support Suppression Pool Spray operation during a Loss of Coolant Accident (LOCA); therefore, this is a violation of TS and reportable per the requirements of 10 CFR 50.73(a)(2)(i)(B).

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Limerick Generating Station, Unit 1	05000352	2018	- 002	- 00

NARRATIVE

I. Unit Conditions Prior to the Event

Limerick Generating Station (LGS) Unit 1 was shut down for refueling outage 1R17 and in Operational Condition (OPCON) 4 at the time of identification. The condition was identified during reverse flushes of Shutdown Cooling piping.

II. Description of the Event

The A LPCI Outboard Injection Valve (HV-051-1F017A) is one of four Low Pressure Coolant Injection (LPCI) outboard injection valves on Unit 1. This valve is a Primary Containment Isolation Valve (PCIV) and a Pressure Isolation Valve (PIV). The A and B LPCI Outboard Isolation Valves have an additional function, in that during a Loss of Coolant Accident (LOCA) one of the two valves will need to be closed to supply water for Suppression Pool Spray.

On 9/8/17, a ground occurred on the Low Pressure Coolant Injection (LPCI) Injection Valve HV-051-1F017A as a result of a steam leak from a nearby Reactor Water Cleanup (RWCU) valve. HV-051-1F017A was determined to be inoperable and was deenergized. The ground alarm cleared following drying of the compartment. The power was restored and the LPCI Injection Valve was returned to operable status. The valve was not stroked because the valve is a PIV.

On 3/27/18, while in OPCON 4, Cold Shutdown, during the performance of reverse flushes of the Shutdown Cooling Piping for refuel outage 1R17, the LPCI Injection Valve HV-051-1F017A was opened. However, the valve failed to stroke fully closed when flushing was complete. The inoperable valve was deenergized and manually closed.

On 4/3/18, a past functionality review concluded that the LPCI Injection Valve was inoperable as a result of the grounding condition.

LGS concluded through evaluation that the A LPCI Outboard Isolation Valve was inoperable from 9/8/17 to 3/27/18 which resulted in the violation of two Technical Specifications (TS) and therefore is reportable per the requirements of 10 CFR 50.73(a)(2)(i)(B).

- TS 3.6.3.a.2 With an inoperable PCIV the penetration must be isolated with a de-energized PCIV within four hours or be in hot shutdown within the next 12 hours and in cold shutdown within the next 24 hours.
- TS 3.6.2.2 With an inoperable Suppression Pool Spray flow path on one of the RHR Loops, return the loop to the operable status within seven days or be in hot shutdown within the next 12 hours and in cold shutdown within the next 24 hours.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

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NARRATIVE

III. Analysis of the Event

When the RWCU steam leak was initially identified it was believed to only affect the humidity in the room. Operations monitored the steam leak via a camera. The issue was evaluated in the Operational Decision-Making Process (ODM) to determine the correct course of action.

On 9/8/17, a ground occurred due to moisture penetrating the limit switch compartment on the motor operator of the A LPCI Outboard Injection Valve. The valve was declared inoperable and isolated. Maintenance opened the motor operator and removed the water, dried the components, and inspected the motor operator. The ground alarm cleared following drying of the compartment. The power was restored and the LPCI Injection Valve was returned to operable status.

IV. Safety Significance

The safety significance was low since the primary containment function was maintained and the injection valve was not opened until the system flushes were performed.

Primary containment was not adversely impacted by this condition because the inboard injection check valve was functional and the RHR system is a qualified closed loop system that is credited as a primary containment boundary.

The LGS long term containment analysis supports the conclusion that the same amount of energy can be removed from the suppression pool following a LOCA whether the exit flow from the RHR Heat Exchanger is injected the reactor vessel or into the suppression pool. Since the LGS procedures allow this method of suppression pool cooling following a LOCA, this function was not lost with the failure of the A LPCI Outboard Injection Valve to close.

Suppression Pool Spray function was maintained because the B Loop of Suppression Pool Spray was maintained available during the time the A Loop LPCI valve was inoperable.

V. Cause of the Event

The cause was determined to be water intrusion from a nearby steam leak that prevented normal control circuit operation. Corrosion to torque switch mechanical parts prevented electrical and mechanical switch makeup on a close demand signal. A contributing cause was identified in that the risk assessment was performed based on incomplete scope of potential impacts. The focus of the risk assessment was on the RWCU system and not the other systems contained in the general area which would have included the Residual Heat Removal (RHR) system injection valve.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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NARRATIVE

VI. Corrective Actions Completed/Planned

HV-051-1F017A was repaired including a replacement of the control circuit hardware.

The station is taking actions to improve risk assessments associated with steam leaks and the potential impact to nearby equipment.

VII. Previous Similar Occurrences

There were no previous similar occurrences in the past 5 years.

VIII. Component Data

System: BO RHR/Low Pressure Coolant Injection System
Component: ISV Isolation Valve
Component number: HV-051-1F017A
Manufacturer: Flowserve Corporation
Model number: 3232-3 DWG
Serial number: 1N292